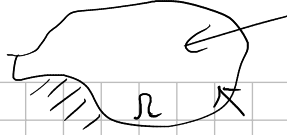


simetría
 $\tau_{ij} = \tau_{ji}$

$$\frac{\partial \tau_{ij}}{\partial x_j} + X_i = 0$$



$$V_i = \frac{\partial \tau_{ji}}{\partial x_j} + X_i = 0 \quad \forall x \in R$$

fuerza másicas o de cuerpo $\equiv 0$

$$\sigma_x = 3x^2 + 4xy - 8y^2 \quad \tau_{xy} = -\frac{1}{2}x^2 - 6xy - 2y^2$$

$$\sigma_y = 2x^2 + xy + 3y^2 \quad \sigma_z = \tau_{xz} = \tau_{yz} = 0$$

$$\tau = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \tau_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \tau_{zz} \end{bmatrix} = \begin{bmatrix} \tau_{xx} & \tau_{xy} & 0 \\ \tau_{xy} & \tau_{yy} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\left\{ \begin{array}{l} \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + X_x = 0 \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + X_y = 0 \\ \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} + X_z = 0 \end{array} \right\} \quad \forall x \in R$$

$$\left\{ \begin{array}{l} \textcircled{1} \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} = 0 \\ \textcircled{2} \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} = 0 \end{array} \right.$$

$$\frac{\partial \tau_{xx}}{\partial x} = 6x + 4y$$

$$\frac{\partial \tau_{yx}}{\partial y} = -x - 6y$$

$$\frac{\partial \tau_{xy}}{\partial x} = x + 6y$$

$$\frac{\partial \tau_{yy}}{\partial y} = -6x - 4y$$

$$\left. \begin{array}{l} \textcircled{1} 6x + 4y - x - 6y = 0 \quad \forall x \in R \\ \textcircled{2} -x - 6y + x + 6y = 0 \quad \forall x \in R \end{array} \right\} \Rightarrow \text{El cuerpo está en equilibrio}$$